

ASHISH KUMAR

Data Engineer | Python | SQL | ETL | Apache Airflow | Agentic AI

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience building Python-based ETL/ELT pipelines, SQL workflows, data-quality validation, and API integrations for a government student-admission platform. Delivered a rank-and-preference-based seat allocation engine processing 1M+ student records using Python, SQL, Apache Airflow, MySQL, ClickHouse, FastAPI, and SQLAlchemy, cutting manual processing time by an estimated 50%+. Skilled in data modeling, workflow orchestration, and schema validation. Currently upskilling in Agentic AI and LLM application development (RAG, MCP, LangGraph) to transition into AI-focused data engineering roles.

PROFESSIONAL EXPERIENCE

Junior Software Developer | Samarth eGov (Ministry of Education, GoI)

June 2026 - Present

- As part of the Data Science Team, developed and maintained a rank-and-preference-based seat allocation engine processing 1M+ student records, including reprocessing logic for displaced candidates across multiple preference lists.
- Engineered merit-ranking logic implementing multi-level tie-breaking rules based on marks, date of birth, and name across 2M+ applicant records, ensuring fair and auditable admissions outcomes.
- Built and orchestrated Python-based ETL pipelines using Apache Airflow to automate data ingestion, transformation, and scheduling, cutting manual data-processing time by an estimated 50%+.
- Implemented schema validation and column-level data-quality checks to proactively identify missing fields and table mismatches, with automated email alerts on pipeline failures that saved an estimated 2-5 hours of manual debugging per week.
- Integrated Strapi and Directus APIs into production data pipelines, built 3-5 internal FastAPI endpoints, and authored ClickHouse queries for internal reporting and analytics.

Data Science Intern | Samarth eGov (Ministry of Education, GoI)

March 2026 - June 2026

- Built foundational data-engineering components for a state-level student admission and seat allocation system used by the Ministry of Education.
- Contributed to seat allocation and merit-ranking logic, MySQL data-extraction workflows, and Apache Airflow DAG development.
- Implemented schema-validation checks to support accurate and reliable processing of admission datasets.

PROJECTS

Adventure Works Sales Analytics | Python | SQL (SQLAlchemy) | Power BI | Pandas

- Built an end-to-end sales analytics pipeline in Python to analyze sales performance and employee productivity metrics.
- Cleaned and transformed raw data using Pandas, handled missing values, and loaded processed data into SQL via SQLAlchemy.
- Designed interactive Power BI dashboards covering sales trends, top performers, and profit margins for stakeholder reporting.

TECHNICAL SKILLS

Programming & Data Engineering: Python, SQL, ETL, ELT, Apache Airflow, Data Modeling, Data Integration, Schema Validation, Multiprocessing

Databases: MySQL, PostgreSQL, ClickHouse, SQLite

Cloud (AWS): S3, Glue, Redshift, Athena, EMR, Lambda, IAM | Also familiar with GCP, Azure

APIs & Development: FastAPI, REST APIs, Strapi, Directus, Streamlit

Data & Analytics: Pandas, NumPy, Scikit-learn, SciPy, Data Cleaning, Data Preprocessing

BI & Visualization: Power BI, Apache Superset, Matplotlib, Seaborn

Tools: Git, GitHub, DataGrip, VS Code, Jupyter Notebook, Excel, Google Sheets

CURRENTLY LEARNING — AGENTIC AI & GENAI

Actively upskilling toward AI/agentic engineering roles: LangChain, LangGraph, OpenAI Agents SDK, MCP, A2A, RAG, Agentic RAG, Context Engineering, Prompt Engineering, Pydantic, Vector Databases, FAISS, Agent Evaluation, Guardrails, LangSmith, AgentOps, Opik, Langfuse, Docker, GitHub Actions, Nginx, OpenAI, Anthropic, Gemini, Groq, Ollama, AWS Bedrock, Azure OpenAI.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering

Bundelkhand University, Jhansi, Uttar Pradesh | August 2019 - June 2023